

Roll No.

--	--	--	--	--	--	--

EVEN END SEMESTER EXAMINATION - 2025

Name of the Course: B. Tech

Semester: VI

Name of the paper: Pressure Die Casting

Paper Code: TME 617

Time: 3 hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory
- (ii) Answer any two sub questions among a, b, & c in each main question
- (iii) Total marks in each main question are twenty
- (iv) Each question carries 10 marks

Q1.	20 marks	
(a)	Define High-Pressure Die Casting (HPDC) and highlight its advantages over conventional casting methods. Describe the specific equipment used in HPDC and suggest one maintenance practice to ensure optimal performance.	CO1
(b)	Define the role of jigs and fixtures in casting processes. Provide an example of a jig or fixture used in casting and explain how it helps maintain dimensional accuracy during casting operations.	
(c)	Identify a casting defect associated with shrinkage and explain its underlying causes. Propose suitable remedies, considering both process modifications and material changes.	
Q2.	20 marks	
(a)	Explain the principle of vacuum casting and discuss its advantages over traditional casting techniques. Mention one specific application where vacuum casting proves particularly effective.	CO2
(b)	Discuss one metallurgical aspect of alloys, such as grain structure or phase composition, and explain its significance in determining the alloy's properties.	
(c)	Explain the purpose of metal treatment in alloy fabrication. Provide one example of a commonly used metal treatment process and describe its effect on the alloy's properties.	
Q3.	20 marks	
(a)	In a sand-casting process with a mold constant of 3 sec/mm ² and a solidification exponent of 2 (i.e., $n = 2$), if the solidification time is tripled for a given unit volume of material, calculate the percentage reduction in the cast surface area	CO3
(b)	Explain the major methods used in riser design within a gating system.	
(c)	Discuss the importance of proper gating system design in minimizing casting defects such as shrinkage and porosity.	
Q4.	20 marks	
(a)	Discuss the role of risers in casting processes. Why is accurate riser design essential, and what key principles should be followed during riser design?	CO4
(b)	A spherical droplet of molten metal with a radius of 8 mm solidifies in 20 seconds. Estimate the solidification time for a similar droplet with a radius of 13 mm.	
(c)	A mold sprue is 69 cm long with a cross-sectional area of 9 cm ² at its base. It feeds a horizontal runner into a mold cavity with a volume of 3100 cm ³ . Determine the time required to fill the mold cavity, assuming $g = 9.8 \text{ m/s}^2$	

Q5.	20 marks	
(a)	Describe the different categories of coatings commonly applied to dies. Discuss their properties and typical applications.*	CO5
(b)	Define die maintenance and explain its significance in the die casting industry. Provide two common maintenance practices for dies and discuss their impact on production efficiency.	
(c)	Define Industry 4.0 and describe its core concepts related to automation, robotics, artificial intelligence, and the Industrial Internet of Things (IoT). Provide one example of how each concept is revolutionizing the die casting industry.	